

Exercise J02

Q1. If $A_1, A_2, A_3, \dots, A_{51}$ are arithmetic means inserted between the numbers a and b , then find the value of $\left(\frac{b + A_{51}}{b - A_{51}}\right) - \left(\frac{A_1 + a}{A_1 - a}\right)$

Ans. 102

Sol. -

Q2. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots$

11. If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to.

Ans. 0

Sol. $a_1 = 15$

$$\frac{a_k + a_{k-2}}{2} = a_{k-1} \text{ for } k = 3, 4, \dots, 11 \text{ के लिए } \Rightarrow a_1, a_2, \dots, a_{11} \text{ are in AP समान्तर श्रेणी में है।}$$

$$a_1 = a = 15$$

$$\frac{a_1^2 + a_2^2 + \dots + a_n^2}{11} = 90$$

$$\Rightarrow \frac{(15)^2 + (15 + d)^2 + \dots + (15 + 10d)^2}{11} = 90$$

$$\Rightarrow 7d^2 + 30d + 27 = 0 \Rightarrow d = -3 \text{ or } -\frac{9}{7}$$

$$\text{Since } 27 - 2a_2 > 0 \Rightarrow a_2 < \frac{27}{2} \Rightarrow d = -3$$

$$\frac{a_1 + a_2 + \dots + a_{11}}{11} = \frac{11}{2} \frac{[30 + 10(-3)]}{11} = 0$$

Q3. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$. For any integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is -

Ans. 3 or 9

Sol. Consider $d \neq 0$ the solution is

$$a_1, a_2, a_3, \dots, a_{100} \rightarrow \text{AP}$$

$$a_1 = 3; S_p = \sum_{i=1}^p a_i \quad 1 \leq n \leq 20$$

$$m = 5n$$

$$\frac{S_m}{S_n} = \frac{\frac{m}{2}[2a_1 + (m-1)d]}{\frac{n}{2}[2a_1 + (n-1)d]}$$

$$\frac{S_m}{S_n} = \frac{5[(2a_1 - d) + 5nd]}{[(2a_1 - d) + nd]}$$

for $\frac{S_m}{S_n}$ to be independent of n

$$\therefore 2a_1 - d = 0 \Rightarrow d = 2a_1 \Rightarrow d = 6$$

$$\Rightarrow a_2 = 9$$

$$\text{If } d = 0 \Rightarrow a_2 = a_1 = 3$$

Q4. The 1st, 2nd and 3rd terms of an arithmetic series are a, b and a² where 'a' is negative. The 1st, 2nd and 3rd terms of a geometric series are a, a² and b find the

(a) value of a and b

(b) sum of infinite geometric series if it exists. If no then find the sum to n terms of the G.P.

(c) sum of the 40 term of the arithmetic series.

Ans. (a) $a = -\frac{1}{2}$, $b = -\frac{1}{8}$; (b) $-\frac{1}{3}$; (c) $\frac{545}{2}$

Sol. -

Q5. For n = 1, 2, 3,, let

$$A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n, \text{ and } B_n = 1 - A_n,$$

Find the smallest natural number n₀ such that B_n > A_n for all n ≥ n₀.

Ans. n₀ = 6

Sol.

$$a_n = \frac{3}{4} - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n, b_n = 1 - a_n = \frac{3}{4} \left[\frac{1 - \left(-\frac{3}{4}\right)^n}{\left(-\frac{3}{4}\right)} \right] = \frac{3}{7}$$

$$\left[1 - \left(-\frac{3}{4}\right)^n \right]$$

$$b_n > a_n \Rightarrow 1 - a_n > a_n$$

$$2a_n < 1 \Rightarrow \frac{6}{7} - \frac{6}{7} \left(-\frac{3}{4}\right)^n < 1$$

$$-\frac{6}{7} \left(-\frac{3}{4}\right)^n < \frac{1}{7} \Rightarrow -3(-3)^n < 2^{2n-1} \Rightarrow (1-)^{n+1} 3^{n+1} < 2^{2n-1}$$

For n even inequality is always valid

For n, odd

; n + 1 → even

$$3^{n+1} < 2^{2n-1}$$

; n = 1, 3² < 2¹, (in valid)

$n = 3, 3^4 < 2^5, 81 < 32$ (in valid) ; min. natural number $n_0 = 6$

Q6. If $x = 1 + a + a^2 + a^3 + \dots$ to ∞ ($|a| < 1$) and $y = 1 + b + b^2 + b^3 + \dots$ to ∞ ($|b| < 1$), then prove that

$$1 + ab + a^2b^2 + a^3b^3 + \dots \infty = \frac{xy}{x + y - 1}.$$

Ans. -

Sol.

$$x = 1 + a + a^2 + \dots = \frac{1}{1 - a}$$

$$y = 1 + b + b^2 + \dots = \frac{1}{1 - b}$$

$$1 + ab + a^2b^2 + \dots = \frac{1}{1 - ab}$$

$$\frac{xy}{x + y - 1} = \frac{\left(\frac{1}{1-a}\right) \left(\frac{1}{1-b}\right)}{\left(\frac{1}{1-a}\right) + \left(\frac{1}{1-b}\right) - 1} = \frac{1}{(1-b) + (1-a) - (1-a)(1-b)}$$

$$= \frac{1}{2 - a - b - (1 - a - b + ab)}$$

$$= \frac{1}{1 - ab} \quad \text{H.P.}$$

Q7. There are 25 trees at equal distances of 5 m in a line with a well, the distance of the well from the nearest tree being 10 m. A gardener waters all the trees separately starting from the well and he returns to the well after watering each tree to get water for the next. Find the total distance, the gardener will cover in order to water all the trees.

Ans. 3500

Sol. -

Q8. The sum of infinite number of terms of a G.P. is 4 and the sum of their cubes is 192. Find the series.

Ans. 6, -3, 3/2,

Sol.

$$\frac{a}{r-1} = 4 \Rightarrow \frac{a^3}{1-r^3} = 192 \Rightarrow \frac{(1-r)^3}{1-r^3} = \frac{192}{(4)^3} \Rightarrow \frac{(1-r)^2}{1+r+r^2} =$$

$$3$$

$$\Rightarrow 1 + r^2 - 2r = 3 + 3r + 3r^2 \Rightarrow 2r^2 + 5r + 2 = 0 \Rightarrow (2r + 1)(r + 2) = 0$$

Q9. The harmonic mean of two numbers is 4. The arithmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.

Ans. 6, 3

Sol. $H = 4 = \frac{2ab}{a+b} \Rightarrow a+b = \frac{ab}{2}$

$$A = \frac{a+b}{2} \quad G = \sqrt{ab}$$

$$2A + G^2 = a+b+ab = 27$$

$$\frac{ab}{2} + ab = 27$$

$$\frac{3ab}{2} = 27$$

$$ab = 18 \Rightarrow a+b = 9$$

$$x^2 - (a+b)x + ab = 0$$

$$x^2 - 9x + 18 = 0$$

$$(x-6)(x-3) = 0$$

$$x = 6, 3$$

Q10. The AM of two numbers exceeds their GM by 15 & HM by 27. Find the numbers.

Ans. 120, 30

Sol. $\frac{a+b}{2} = \sqrt{ab} + 15 \quad \frac{a+b}{2} = \frac{2ab}{a+b} + 27$

$$2\sqrt{ab} = (a+b) - 30 \quad (a+b)^2 = (2\sqrt{ab})^2 + 54(a+b)$$

$$(a+b)^2 = [(a+b) - 30]^2 + 54(a+b)$$

$$\text{Let } a+b = x$$

$$x^2 = x^2 + 900 - 60x + 54x$$

$$6x = 150$$

$$x = 150$$

$$a+b=150 \Rightarrow \frac{150}{2}15 = \sqrt{ab}$$

$$60 = \sqrt{ab}$$

$$ab = 3600$$

$$y^2 + \{-(a+b)\}y + ab = 0$$

$$y^2 - 150y + 3600 = 0$$

$$(y-120)(y-30) = 0$$

$$y = 30, 120$$

Q11. If a, b, c are positive real numbers, then prove that $[(1+a)(1+b)(1+c)]^7 > 7^7 a^4 b^4 c^4$.

Ans. (a) C, (b) $n=7$

Sol. -

Q12. The minimum value of the sum of real numbers a^{-5} , a^{-4} , $3a^{-3}$, 1, a^8 and a^{10} with $a > 0$ is

Ans. 8

Sol. As $a > 0$ and all the given terms are positive hence considering A.M. $\geq$ G.M. for given numbers :

$$\frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + a^8 + a^{10}}{7} \geq (a^{-5} \cdot a^{-4} \cdot a^{-3} \cdot a^{-3} \cdot a^{-3} \cdot a^8 \cdot a^{10})^{1/7}$$
$$\Rightarrow \frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^{-3} + a^8 + a^{10}}{7} \geq 7$$

$$(a^{-5} + a^{-4} + 3a^{-3} + a^8 + a^{10})_{\min} = 7$$

where $a^{-5} + a^{-4} = a^{-3} = a^8 = a^{10}$ i.e. $a = 1$

$$\Rightarrow (a^{-5} + a^{-4} + 3a^{-3} + a^8 + a^{10} + 1)_{\min} = 8 \text{ when } a = 1$$

Q13. If the 10th term of an HP is 21 and 21st term of the same HP is 10, then find the 210th term.

Ans. (a) 6, 3; (b) 120, 30

Sol. -

Q14. Let ' σ ' denotes the sum of the infinite series $\sum_{n=1}^{\infty} \left(\frac{n^2 + 2n + 3}{2^n} \right)$.

Compute the value of $(1^3 + 2^3 + 3^3 + \dots + \sigma^3)$.

Ans. 8281

Sol. -

Q15. If total number of runs scored in n matches is $\left(\frac{n+1}{4} \right) (2^{n+1} - n - 2)$ where $n > 1$, and the runs scored in the k^{th} match are given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$. Find n .

Ans. $n = 7$

Sol. $T_k = k \cdot 2^{n+1-k}$ and $S_n = \left(\frac{n+1}{4} \right) (2^{n+1} - n - 2)$

Now, $S_n = \sum_{k=1}^n k \cdot 2^{-k} = 2^{n+1} \sum_{k=1}^n k \cdot 2^{-k}$; $S_b = 2^{n+1} \cdot 2 \cdot \left[1 - \frac{1}{2^n} - \frac{n}{2^{n+1}} \right]$ (sum of A.G.P.)

Q16. If $S_1, S_2, S_3, \dots, S_n, \dots$ are the sums of infinite geometric series whose first terms are 1, 2, 3, ... n , ... and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n+1}, \dots$ respectively, then

find the value of $\sum_{r=1}^{2n-1} S_r^2$.

Ans. $\frac{n(2n+1)(4n+1)}{3} - 1$

Sol. -

Q17. Find the nth term and the sum to n terms of the sequence:

(i) $1 + 5 + 13 + 29 + 61 + \dots$ (ii) $6 + 13 + 22 + 33 + \dots$

Ans. (i) $2^{n+1} - 3$; $2^{n+2} - 4 - 3n$ (ii) $n^2 + 4n + 1$; $(1/6)n(n+1)(2n+13) + n$

Sol. (i) $S = 1 + 5 + 13 + 29 + 61 + \dots T_n$

$$S = 1 + 5 + 13 + 29 + \dots T_n$$

$$T_n = 1 + (2^{n-1} - 1) = 2^{n+1} - 3$$

$$S_n = \sum (2^{n+1} - 1) = (2^2 + 2^3 + \dots 2^{n+1}) - 3n$$

$$= 2^2 \left(\frac{2^n - 1}{2 - 1} \right) - 3n = 2^{n+2} - 4 - 3n$$

(ii) $S = 6 + 13 + 22 + 33 + \dots T_n$

$$S = 6 + 13 + 22 + \dots T_n$$

$$T_n = 6 + 7 + 9 + 11 \dots$$

$$= 1 + 5 + 7 + 9 + 11 \dots$$

$$= 1 + \frac{n}{2} (10(n-1)2) = n^2 + 4n + 1$$

$$S_n = \sum T_n = \frac{1}{6} n(n+1)(2n+13) + n$$

Q18. If the sum

$$\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \sqrt{1 + \frac{1}{3^2} + \frac{1}{4^2}} + \dots + \sqrt{1 + \frac{1}{(1999)^2} + \frac{1}{(2000)^2}}$$

equal to $n - \frac{1}{n}$ where $n \in \mathbb{N}$. Find n.

Ans. $n = 2000$

Sol. -

Q19. The series of natural numbers is divided into groups (1); (2, 3, 4); (5, 6, 7, 8, 9); ... and so on. Show that the sum of the numbers in the nth group is $(n-1)^3 + n^3$.

Ans. -

Sol. -

Q20. Sum the following series to n terms and to infinity :

(i) $\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$ (ii) $\sum_{r=1}^n r(r+1)(r+2)(r+3)$

(iii) $\sum_{r=1}^n \frac{1}{4r^2 - 1}$ (iv) $\frac{1}{2.4} + \frac{1.3}{2.4.6} + \frac{1.3.5}{2.4.6.8} + \dots$

Ans. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}]$; $s_\infty = 1/24$ (ii) $(1/5) n(n+1)(n+2)(n+3)(n+4)$

(iii) $n/(2n+1)$ (iv) $S_n = \left[\frac{1}{2} - \frac{1.3.5 \dots (2n-1)(2n+1)}{2.4.6 \dots (2n)(2n+2)} \right]$; $S_\infty = \frac{1}{2}$

Sol. (i) $\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$

$$T_n = \frac{1}{(3n-2)(3n+1)(3n+4)}$$

$$= \frac{1}{3n+1} \left(\frac{1}{3n-2} - \frac{1}{(3n+4)} \right) \frac{1}{6}$$

$$= \left(\frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)} \right) \frac{1}{6}$$

$$S_n = \frac{1}{6} \left[\frac{1}{1.4} - \frac{1}{4.7} + \frac{1}{4.7} - \frac{1}{7.10} + \frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)} \right]$$

$$\frac{1}{6} \left[\frac{1}{4} - \frac{1}{(3n+1)(3n+4)} \right]$$

now $S_\infty = \frac{1}{24}$

(ii) $T_n = n(n+1)(n+2)(n+3)$

$$= \frac{1}{5} [n(n+1)(n+2)(n+3)((n+3)-(n-1))]$$

$$= \frac{1}{5} [n(n+1)(n+2)(n+3)(n+4-(n-1)(n)(n+2)(n+3))]$$

$$\sum T_n = \left(\frac{1}{5} \right) n(n+1)(n+2)(n+3)(n+4)$$

(iii) $T_n = \frac{1}{4n^2-1} = \frac{1}{2} \left[\frac{1}{2n-1} - \frac{1}{2n+1} \right]$

$$S_n = \frac{1}{2} \left[1 - \frac{1}{2n+1} \right]$$

$$S_\infty = \frac{1}{2}$$

Q21. If $\sum_{1}^{20} r^3 = a$, $\sum_{1}^{20} r^2 = b$ then sum of products of 1, 2, 3, 4 20 taking two at a time is :

Ans. $\frac{a-b}{2}$

Sol. -

Paragraph

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

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Q22. Which one of the following statements is correct?

- (A) $G_1 > G_2 > G_3 > \dots$ (B) $G_1 < G_2 < G_3 < \dots$ (C) $G_1 = G_2 = G_3 = \dots$
(D) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

Ans. C

Sol. $A_1 = \frac{a+b}{2}$; $G_1 = \sqrt{ab}$; $H_1 = \frac{2ab}{a+b}$

$$A_n = \frac{A_{n-1} + H_{n-1}}{2}; G_n = \sqrt{A_{n-1}H_{n-1}}$$

$$H_n = \frac{2A_{n-1}H_{n-1}}{A_{n-1} + H_{n-1}}$$

We know that $G^2 = AH$

So clearly $G_1 = G_2 = G_3 = \dots G_n = \sqrt{ab}$

Paragraph

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

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Q23. Which one of the following statements is correct?

(A) $A_1 > A_2 > A_3 > \dots$ (B) $A_1 < A_2 < A_3 < \dots$

(C) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$

(D) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

Ans. A

Sol. A_2 is AM of A_1, H_1 and $A_1 > H_1$

$$\Rightarrow A_1 > A_2 > H_1$$

A_3 is A.M. of A_2, H_2

$$\Rightarrow A_2 > A_3 > H_2$$

$$\therefore A_1 > A_2 > A_3 \dots\dots\dots$$

Paragraph

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

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Q24. Which one of the following statements is correct?

(A) $H_1 > H_2 > H_3 > \dots$ (B) $H_1 < H_2 < H_3 < \dots$

(C) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$

(D) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$

Ans. B

Sol. as above

$$A_1 > H_2 > H_1$$

$$A_2 > H_3 > H_2$$

.....

$$\text{so } \dots\dots\dots H_3 < H_2 < H_1 \dots\dots\dots$$

$$\Rightarrow H_1 < H_2 < H_3 \dots\dots\dots$$

Q25. In an A.P., let T_r denote r th term from beginning, $T_p = \frac{1}{q(p+q)}$, $T_q = \frac{1}{p(p+q)}$, then :

- (A) $T_1 =$ common difference (B) $T_{p+q} = \frac{1}{pq}$ (C) $T_{pq} = \frac{1}{p+q}$ (D) $T_{p+q} = \frac{1}{p^2q^2}$

Ans. A, B, C

Sol. -

Q26. All roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are in G.P. If the sum of their reciprocals is 10, then d can be equal to :

- (A) 32 (B) -32 (C) $\frac{1}{32}$ (D) $-\frac{1}{32}$

Ans. A, B

Sol. -

Q27. For the A.P. given by $a_1, a_2, \dots, a_n, \dots$, the equations satisfied are

- (A) $a_1 + 2a_2 + a_3 = 0$ (B) $a_1 - 2a_2 + a_3 = 0$ (C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$
(D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$

Ans. D, B

Sol. -

Q28. If sum of the infinite G.P., $p, 1, \frac{1}{p}, \frac{1}{p^2}, \frac{1}{p^3}, \dots$ is $\frac{9}{2}$, the value of p is

- (A) 3 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{1}{3}$

Ans. A, C

Sol. -

Q29. If the arithmetic mean of two positive numbers a & b ($a > b$) is twice their geometric mean, then $a : b$ is:

- (A) $2 + \sqrt{3} : 2 - \sqrt{3}$ (B) $7 + 4\sqrt{3} : 1$ (C) $1 : 7 - 4\sqrt{3}$ (D) $2 : \sqrt{3}$

Ans. A, B, C

Sol. $\frac{\frac{a+b}{2}}{\sqrt{ab}} = \frac{1}{2} \Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{1}{2}$ use compendo and dividendo rule

$$\Rightarrow \frac{a+b+2\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}} = \frac{\sqrt{3}}{1} \Rightarrow \frac{2\sqrt{a}}{2\sqrt{b}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$\Rightarrow \frac{a}{b} = \frac{3+1+2\sqrt{ab}}{a+b-2\sqrt{ab}} \Rightarrow \frac{a}{b} = \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}} = \frac{(2+\sqrt{3})(2+\sqrt{3})}{4-3} = 7+4\sqrt{3} \text{ Ans.}$$

Q30. If $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then

- (A) $a + c = b + d$ (B) $e = 0$ (C) $a, b - 2/3, c - 1$ are in A.P. (D) c/a is an integer

Ans. A, B, C, D

Sol. $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$

$$\sum_{r=1}^n (r^2 + r)(2r+3) = \sum_{r=1}^n (2r^3 + 5r^2 + 3r) = 2 \cdot \frac{n^2(n+1)^2}{4} + 5 \frac{n(n+1)(2n+1)}{6} + 3 \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2} \left[n(n+1) + \frac{5}{3}(2n+1) + 3 \right] = \frac{n(n+1)}{2} [6n^2 + 26n + 28]$$

$$= \frac{6}{12}; \quad b = \frac{32}{12}; \quad c = \frac{54}{12}; \quad d = \frac{28}{12} = \frac{7}{3}; \quad e = 0$$

so $a + c = b + d$

$$b - \frac{2}{3} = \frac{32}{12} - \frac{2}{3} = \frac{24}{12}; \quad c - 1 = \frac{42}{12}$$

so $a, b - 2/3, c - 1$ are in A.P. & $\frac{c}{a} = \frac{54}{6} = 9$ is an integer

Q31. If the first two terms of a progression are 8 and 4 respectively, then correct statement(s) is/are:

- (A) if third term is 2, then the terms are in G. P.
 (B) if third term is 0, then the terms are in A. P.
 (C) if third term is $\frac{8}{3}$, then the terms are in H. P.
 (D) if the third term is 3, then terms are in A. P.

Ans. A, B, C

Sol. -

Q32. If $40 + \frac{155}{4} + \frac{150}{4} + \frac{145}{4} + \dots + n$ terms = 625 then n can be

- (A) 24 (B) 25 (C) 40 (D) 37

Ans. B, C

Sol. -

Q33.

Column – I

Column – II

(A) If $\log_5 2, \log_5(2^x - 5)$ and $\log_5(2^x - 7/2)$ are in A.P., then value of $2x$ is equal to

(P) 6

(B) Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, (Q) 9
 then is $\frac{S_{3n}}{S_n}$

(C) Sum of infinite series $4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^2} + \dots$ is (R) 3

(D) The length, breadth, height of a rectangular box are in G.P. The (S) 1
 volume is 27, the total surface area is 78. Then the length is

(A) (A) P, (B) P, (C) Q, (D) Q, S (B) (A) P, (B) P, (C) Q, P (D) S

(C) (A) Q, (B) P, (C) Q, (D) S (D) (A) P, (B) S, (C) Q, P (D) Q, S

Ans. 5ee5c569f98f7e49639cff0d

Sol. -

Q34. **Column – I** **Column – II**

(A) If $\log_x y$, $\log_y x$, $\log_y z$ are in G.P., $xyz = 64$ and x^3, y^3, z^3 (P) 2
 are in A.P., then $\frac{3x}{y} =$

(B) The value of $2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \dots =$ (Q) 1

(C) If x, y, z are in A.P., then $(x + 2y - z)(2y + z - x)(z + x - y) =$ (R) 3
 $kxyz$, where $k \in \mathbb{N}$, then $k =$ (S) 4

(A) (A) R, (B) Q, (C) S, (B) (A) R, (B) P, (C) S,

(C) (A) R, (B) Q, (C) P, S (D) (A) R, Q (B) Q, (C) P,

Ans. 5ee5c8fdf98f7e49639cff33

Sol. (A) a, x, y, z, b in A.P. & a, x, y, z, b in H.P.

$\frac{1}{b}, \frac{1}{z}, \frac{1}{y}, \frac{1}{a}$ in A.P. ; $a, \frac{ab}{z}, \frac{ab}{y}, \frac{ab}{x}, b$ in A.P.

$ab = xz = y^2 = zx = bz \Rightarrow xz \cdot y^2 \cdot zx = (ab)(ab)(ab) \Rightarrow (xyz)$
 $(xyz) = a^3 b^3$

$ab = 3 \Rightarrow a = 1, b = 3$ or $a = 3, b = 1$

(B) $S_\infty = 2^{1/14} \cdot 4^{1/8} \cdot 8^{1/16} \dots \infty$; $S_\infty = 2^{1/14} \cdot 2^{2/8} \cdot 2^{3/16} \dots \infty$

$S_\infty = 2^{1/14 + 2/8 + 3/16} \dots \infty = 2^{s'_\infty}$

Let $S'_\infty = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots$

$= \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots + \frac{n}{2^{n+1}} \dots \dots \dots (i)$

$\frac{S'_n}{2} = \frac{1}{8} + \frac{2}{16} + \dots + \frac{n-1}{2^{n+2}} \dots \dots \dots (ii)$

(i) – (ii) we get

$$\frac{S'_n}{2} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \frac{1}{2^{n+1}} - \frac{n}{2^{n+2}} \Rightarrow \frac{S'_n}{2} = \frac{1}{4} \left(\frac{1 - (1/2)^n}{1 - 1/2} \right) - \frac{1}{2^{n+2}}$$

$$S'_n = \frac{2.2}{4} \left(1 - \left(\frac{1}{2} \right)^n \right) - \frac{2n}{2^{n+2}} \Rightarrow S'_\infty = 1 \quad \therefore S_\infty = 2^{s'_\infty} = 2$$

(C) x, y, z are in A.P. $y = \frac{x+z}{2}$, or $2y = x+z$

$$(x+2y-z)(x+z+z-x)(z+x-y) = (x+(x+z)-z)(x+z+z-x)(2y-y) = 2x.$$

$$2z \cdot y = 4xyz$$

$$\therefore k = 4$$

$$(D) d = \frac{31-1}{m+1} = \frac{30}{m+1}; \quad \frac{A_7}{A_{m-1}} = \frac{1+7\frac{30}{m+1}}{1+(m-1)\frac{30}{m+1}} = \frac{5}{9}$$

$$\Rightarrow \frac{m+211}{31m-29} = \frac{5}{9} \Rightarrow 146m = 2044 \Rightarrow m = 14 \quad \therefore \frac{m}{7} = 2$$

Q35. Match the series in Column I with their sum in Column II

Column I

(A) $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots$ to infinity equals

(B) $\frac{2^3-1}{2^3+1} \times \frac{3^3-1}{3^3+1} \times \frac{4^3-1}{4^3+1} \dots$ to infinity equals

(C) $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$ to infinity equals

(D) $\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \frac{1}{3.4.5.6} + \dots$ to infinity equals

Column II

(P) $\frac{1}{4}$

(Q) $\frac{1}{2}$

(R) $\frac{2}{3}$

(S) $\frac{1}{18}$

(A) (A) Q, (B) R, (C) S, (D) P (B) (A) R, (B) Q, (C) P, (D) S

(C) (A) Q, (B) R, (C) P, (D) S (D) (A) Q, (B) R, (C) S, (D) R

Ans. 5ee5cc1b56a132497f264154

Sol. -